

MATHEMATICS

TARGET : JEE- Advanced

CAPS-16

INDEFINITE INTEGRATION

SCQ (Single Correct Type) :

1. The value of $\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals:
- (A) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x + C$ (B) $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$
- (C) $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + C$ (D) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + x + C$
2. The value of $2 \int \sin x \cdot \operatorname{cosec} 4x dx$ is equal to
- (A) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ (B) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| + \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$
- (C) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2}\sin x}{1+\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ (D) none of these
3. The values of a and b which satisfy $\int \frac{2\sin 2x - \cos x}{4 - \cos^2 x - 4\sin x} dx = a \log |\sin x - 1| + b \log |\sin x - 3| + c$ are
- (A) $-\frac{3}{2}, \frac{11}{2}$ (B) $\frac{3}{2}, \frac{11}{2}$ (C) $\frac{3}{2}, -\frac{11}{2}$ (D) $-\frac{3}{2}, -\frac{11}{2}$
4. Let $\int \frac{g(x)((1+x)\ln x - 5)}{x(\ln x)^6} dx = \frac{g(x)}{(\ln x)^5} + c, g(1) = e$ and $g: (0, e) \rightarrow (0, \infty)$. If $g^n(x)$ denotes the n^{th} derivative of $g(x)$, then the value of $g^{2012}(1)$ is _____.
- (A) $2011g(1)$ (B) $2012g(1)$ (C) $2013g(1)$ (D) $2014g(1)$
5. $\int e^{\sin x} \left(\frac{x \cos^3 x - \sin x}{\cos^2 x} \right) dx =$
- (A) $e^{\sin x} (x + \sec x) + c$ (B) $e^{\sin x} (1 + \sec x) + c$
- (C) $e^{\sin x} (1 - \sec x) + c$ (D) $e^{\sin x} (x - \sec x) + c$
6. If $\int \frac{xdx}{2012\sqrt{(1+x^2)^{1012}(2+x^2)^{3012}}} = \frac{\alpha}{\beta} (1-f(x))^{\frac{\beta}{2\alpha}} + k$ then which is true
- (A) $\alpha = 503; \beta = 500, f(\sqrt{2}) = \frac{1}{\beta - \alpha}$ (B) $\alpha = 503; \beta = 250, f(\sqrt{2}) = \frac{1}{\alpha - \beta}$
- (C) $\alpha = 503; \beta = 500, f(1) = \frac{1}{\alpha - \beta}$ (D) $\alpha = 503; \beta = 225, f(\sqrt{3}) = \frac{1}{\alpha - \beta}$



7. If $\int \frac{\sec^2 x - 2010}{\sin^{2010} x} dx = \frac{P(x)}{\sin^{2010} x} + C$, then value of $P\left(\frac{\pi}{3}\right)$ is ____.
- (A) 0 (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) None of these
8. If $g(x) = \frac{f(x)}{(x-a)(x-b)(x-c)}$, where $f(x)$ is a polynomial of degree < 3 , then ____.
- (A) $\int g(x) dx = \begin{vmatrix} 1 & a & f(a)\log|x-a| \\ 1 & b & f(b)\log|x-b| \\ 1 & c & f(c)\log|x-c| \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + k$
- (B) $\frac{d}{dx} g(x) = \begin{vmatrix} 1 & a & f(a)\log(x-a)^{-2} \\ 1 & b & f(b)\log(x-b)^{-2} \\ 1 & c & f(c)\log(x-c)^{-2} \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$
- (C) $\int g(x) dx = \begin{vmatrix} 1 & a & f(a)\log|x-a| \\ 1 & b & f(b)\log|x-b| \\ 1 & c & f(c)\log|x-c| \end{vmatrix} \div \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} + k$
- (D) None of these
9. If $y = 2 \cot^{-1}\left(\frac{x}{2}\right) + x$, then $\int \frac{x^2 dx}{(x^2 - 4)\sin x + 4x \cos x}$
- (A) $\log|\operatorname{cosec} 2y + \cot y| + c$ (B) $\log|\operatorname{cosec} 2y - \cot 2y| + c$
- (C) $\log|\operatorname{cosec} y + \cot y| + c$ (D) $\log|\operatorname{cosec} y - \cot y| + c$

Comprehension Type Question:

Comprehension # 1

In calculating a number of integrals we had to use the method of integration by parts several times in succession. The result could be obtained more rapidly and in a more concise form by using the so-called generalized formula for integration by parts

$$\int u(x) v(x) dx = u(x) v_1(x) - u'(x) v_2(x) + u''(x) v_3(x) + \dots + (-1)^{n-1} u^{n-1}(x) v_n(x) - (-1)^{n-1} \int u^n(x) v_n(x) dx$$

where $v_1(x) = \int v(x) dx$, $v_2(x) = \int v_1(x) dx$, ..., $v_n(x) = \int v_{n-1}(x) dx$

Of course, we assume that all derivatives and integrals appearing in this formula exist. The use of the generalized formula for integration by parts is especially useful when calculating $\int P_n(x) Q(x) dx$, where $P_n(x)$ is polynomial of degree n and the factor $Q(x)$ is such that it can be integrated successively $n + 1$ times.

10. If $\int (x^3 - 2x^2 + 3x - 1) \cos 2x \, dx = \frac{\sin 2x}{4} u(x) + \frac{\cos 2x}{8} v(x) + C$ then
- (A) $u(x) = x^3 - 4x^2 + 3x$ (B) $u(x) = 2x^3 - 4x^2 + 3x$
 (C) $v(x) = 3x^2 - 4x + 3$ (D) $v(x) = 6x^2 - 8x$
11. If $\int (3x^2 + x - 2) \sin^2(3x + 1) \, dx = -\frac{u(x)}{72} \sin(6x + 2) + \frac{v(x)}{72} \cos(6x + 2) + \frac{1}{2}x^3 + \frac{1}{4}x^2 - x + C$
- (A) $u(x) = 3x^2 + 6x - 13$ (B) $u(x) = 18x^2 + 2x - 13$
 (C) $v(x) = 3x + 1$ (D) $v(x) = -(6x + 1)$

Numerical based Questions :

12. If $\int \frac{\sqrt{4+x^2}}{x^6} \, dx = \frac{(a+x^2)^{3/2} \cdot (x^2-b)}{120x^5} + C$ then $a + b$ equals to :
13. Let $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2} \, dx$ and $g(0) = 0$ then value of $32g\left(\frac{\pi}{2}\right)$ is.
14. If $f(x) = \sqrt{x-1}$; $g(x) = e^x$ and $\int f(g(x)) \, dx = A f(g(x)) + B \tan^{-1}(f(g(x))) + C$ then $A^3 + B^2$ equals
15. If $\int \frac{1+x\cos x}{x(1-x^2e^{2\sin x})} \, dx = k \ln \sqrt{\frac{x^2e^{2\sin x}}{1-x^2e^{2\sin x}}} + C$ then k is equal to :
16. Let $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}} \, dx = \sqrt{m} \tan^{-1}\left(\sqrt{\frac{f(x)-g(x)}{ng(x)}}\right) + C$,

where $m, n \in \mathbb{N}$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$.

Matrix Match Type :

17. Let $I = \int \frac{e^x}{e^{4x} + 1} \, dx$ and $J = \int \frac{e^{-x}}{e^{-4x} + 1} \, dx$

Then for any arbitrary constant C, match the following

Column-I

- (A) I
 (B) $J + I$
 (C) $J - I$

Column-II

- (P) $\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) + C$
 (Q) $\frac{1}{2\sqrt{2}} \ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right) + C$
 (R) $\frac{1}{2\sqrt{2}} \left[\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) - \frac{1}{2} \ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right) \right] + C$
 (S) $\frac{1}{2\sqrt{2}} \left[\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) + \frac{1}{2} \ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right) \right] + C$

18. Column-I

(A) $\int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$

(B) $\int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$

(C) $\int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$

(where C is the constant of integration.)

Column-II

(P) $\frac{1}{6} \left(\frac{1}{x^{-3}} + \frac{3}{x^{-2}} \right)^{\frac{1}{2}} + C$

(Q) $\frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$

(R) $\frac{-1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$

(S) $x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

Subjective Type Questions :

19. Evaluate : $\int \cos x \cdot e^x \cdot x^2 dx$

20. Evaluate : $\tan^{-1} x \cdot \ln(1 + x^2) dx$.

21. Evaluate : $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1 - x^n) \sqrt{1 - x^{2n}}} dx$

22. If $\int \frac{x \cos \alpha + 1}{(x^2 + 2x \cos \alpha + 1)^{3/2}} dx = \frac{f(x)}{\sqrt{g(x)}} + C$ then find f(x) and g(x).

23. $\int \frac{dx}{(\sin^2 x + 2 \cos^2 x)^2} dx$

24. Evaluate $\int \frac{(x - \sin x)^{3/2}}{\sqrt{x}} \left\{ \frac{6x^2 \sin^2 \frac{x}{2}}{x - \sin x} + 3x \right\} dx$

ANSWER KEY OF CAPS-16

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|---------|---------------------------|-------------------------|----------|---------|
| 1. (A) | 2. (A) | 3. (A) | 4. (C) | 5. (D) |
| 6. (C) | 7. (C) | 8. (A) | 9. (D) | 10. (B) |
| 11. (D) | 12. (10) | 13. (16) | 14. (12) | 15. (1) |
| 16. (8) | 17. (A) R ; (B) P ; (C) Q | 18. (A) R, (B) S, (C) Q | | |

19. $\frac{1}{2} e^x [(x^2 - 1) \cos x + (x - 1)^2 \cdot \sin x] + C$

20. $x \tan^{-1} x \cdot \ln(1 + x^2) + (\tan^{-1} x)^2 - 2x \tan^{-1} x + \ln(1 + x^2) - \left(\ln \sqrt{1 + x^2} \right)^2 + C$

21. $e^x \sqrt{\frac{1 + x^n}{1 - x^n}} + C$

22. $x; x^2 + 2x \cos \alpha + 1$

23. $\frac{3}{8\sqrt{2}} \tan^{-1} \left(\frac{t^2 x - 2}{2\sqrt{2} \tan x} \right) - \frac{\tan x}{4(\tan^2 x + 2)} + C$

24. $I = 3 \int \sqrt{t} dt = 3 \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + c = 2 \left(x^2 - x \sin x \right)^{\frac{3}{2}} + c$